

지구를 위해 행동하라

교과서: 영어 I

출판사: 동아(박용예)

단원: Lesson 3

문항: 20문항 / 20분

이름: _____

학교: _____

학년: _____

점수: _____ / 100

[배점] 쉬움 4문항=12점 | 보통 9문항=45점 | 어려움 7문항=43점 = 총 100점 (80점 이상 통과)

01 예상문제 — 지문을 읽으며 문제를 풀어보세요!

01 어법 · Grammar

01 어법 ㉠~㉥ 보통

[5점]

다음 글의 밑줄 친 ㉠~㉥ 중 어법상 어색한 것을 고르시오.

Whales are amazing creatures. They are enormous and live incredibly long lives. The largest blue whales can grow as long as 30 meters and weigh over 200 tons, and some whales have been ㉠ known to live for more than 100 years. These giants do extremely ㉡ beneficial things for the environment throughout their long lives and even after death. According to scientists, a single whale can absorb about 550 kilograms of CO₂ per year, while a tree can absorb only 22. Moreover, after they die, whales sink to the ocean floor with all the CO₂ they ㉢ absorbing during their lives. Their bodies then become food and habitats for deep-sea creatures. This process ㉣ is called a whale fall. Whale falls permanently ㉤ bury about 145,000 tons of CO₂ under the sea every year.

- ① ㉠ known
- ② ㉡ beneficial
- ③ ㉢ absorbing
- ④ ㉣ is called
- ⑤ ㉤ bury

02 어법 A/B/C 어려움

[6점]

다음 글의 괄호 (A), (B), (C) 안에서 어법에 맞는 표현으로 가장 적절한 것을 고르시오.

Whales dive deep into the ocean to feed on prey such as krill and small fish, and they occasionally swim up to the surface to breathe. When they near the surface, they release an enormous amount of waste. This waste is rich in nutrients (A) [that / what] are essential for creatures living in that area. Specifically, it creates the perfect growing conditions for some microscopic creatures (B) [floating / floated] at the surface of the ocean. These creatures, called phytoplankton, may look tiny and insignificant, but when (C) [taking / taken] together, they play a significant role in reducing CO₂.

- ① (A) that - (B) floating - (C) taken
- ② (A) what - (B) floating - (C) taken
- ③ (A) that - (B) floated - (C) taking
- ④ (A) what - (B) floated - (C) taken
- ⑤ (A) that - (B) floating - (C) taking

03 어법 ㉠~㉣ 어려움

[6점]

다음 글의 밑줄 친 ㉠~㉣ 중 어법상 어색한 것을 고르시오.

Despite the important roles of whales in our environment, their lives are at serious risk. The number of whales ㉠ had dramatically dropped until killing them for commercial purposes was officially banned in 1986. It is estimated that the global whale population ㉡ fell from 4 to 5 million to just over 1 million during the decades before the ban. Today, whales are still threatened by various man-made dangers, such as fishing nets, ship strikes, and microplastic pollution in the ocean. It is important ㉢ to recognize that whales are valuable partners in our fight against the climate crisis and that ㉣ protecting them is an effective and cost-efficient way to address this global challenge.

- ① ㉠ had dramatically dropped
- ② ㉡ fell
- ③ ㉢ to recognize
- ④ ㉣ protecting

A

02 어휘 · Vocabulary

다음 글의 괄호 (A), (B), (C) 안에서 문맥에 맞는 낱말로 가장 적절한 것을 고르시오.

Over the past two centuries, human activities have generated a (A) [significant / insignificant] amount of carbon dioxide. This gas itself is not dangerous when we encounter it in everyday situations. However, excessive amounts of this gas in the atmosphere have contributed to the greenhouse effect, resulting in a (B) [gradual / sudden] increase in the Earth's surface temperature. To make matters worse, this increase has sped up in recent years and is seriously (C) [threatening / protecting] life on our planet.

- ① (A) significant – (B) gradual – (C) threatening
- ② (A) insignificant – (B) gradual – (C) threatening
- ③ (A) significant – (B) sudden – (C) threatening
- ④ (A) significant – (B) gradual – (C) protecting
- ⑤ (A) insignificant – (B) sudden – (C) protecting

다음 글의 밑줄 친 ①~⑤ 중 문맥상 낱말의 쓰임이 적절하지 않은 것을 고르시오.

Whales dive deep into the ocean to feed on prey such as krill and small fish, and they occasionally swim up to the surface to breathe. When they near the surface, they release an enormous amount of waste. This waste is rich in nutrients that are ① essential for creatures living in that area. Specifically, it creates the perfect growing conditions for some ② microscopic creatures floating at the surface of the ocean. These creatures, called phytoplankton, may look tiny and ③ significant, but when taken together, they play a ④ significant role in reducing CO₂. They absorb about 40% of the carbon in the atmosphere and produce about 50% of the atmosphere's oxygen through photosynthesis, a process that creates energy using carbon, water, and ⑤ light.

- ① ① essential
- ② ② microscopic
- ③ ③ significant
- ④ ④ significant
- ⑤ ⑤ light

다음 글의 밑줄 친 ①~⑤ 중 문맥상 낱말의 쓰임이 적절하지 않은 것을 고르시오.

Whales spread nutrients in the ocean not only vertically but also ① horizontally through a process called the great whale conveyor belt. These giant mammals ② migrate over long distances at certain times of the year. For example, in the summer, humpback whales tend to stay near Alaska, where the water is cold but rich in nutrients. As winter approaches, they travel up to 4,800 kilometers to breed in the warm waters near Hawaii. Nutrients are relatively ③ abundant in warm waters, which makes it difficult for phytoplankton to survive. Fortunately, when the whales arrive in Hawaii after eating their fill in Alaska, they release a large amount of ④ nutrient-rich waste. These nutrients support the life of many marine animals and, importantly, ⑤ encourage the growth of phytoplankton even in warm waters.

- ① ① horizontally
- ② ② migrate
- ③ ③ abundant
- ④ ④ nutrient-rich
- ⑤ ⑤ encourage

B

03 대의 · Main Idea

다음 글의 주제로 가장 적절한 것을 고르시오.

In addition to their direct absorption and burial of CO₂ through whale falls, whales also make a significant indirect contribution to reducing CO₂ through an act called a whale pump. This refers to the vertical movements of whales that help circulate nutrients across different depths of the ocean. When they near the surface, they release an enormous amount of waste rich in nutrients that are essential for creatures living in that area. These creatures, called phytoplankton, absorb about 40% of the carbon in the atmosphere and produce about 50% of the atmosphere's oxygen through photosynthesis.

- ① the role of whale pumps in reducing CO₂ through nutrient circulation
- ② how whales absorb CO₂ directly from the atmosphere
- ③ the importance of phytoplankton in deep-sea ecosystems
- ④ various methods to protect endangered whale species
- ⑤ the harmful effects of CO₂ on marine life

다음 글의 제목으로 가장 적절한 것을 고르시오.

While many high-tech solutions such as carbon capture have been proposed, they are often too expensive to be used widely. Fortunately, there are also low-tech, economical, and even long-lasting solutions available. According to scientists, a single whale can absorb about 550 kilograms of CO₂ per year, while a tree can absorb only 22. Moreover, after they die, whales sink to the ocean floor with all the CO₂ they absorbed during their lives. Whale falls permanently bury about 145,000 tons of CO₂ under the sea every year. This achievement is basically what expensive carbon capture technologies aim to do.

- ① Whales: Nature's Cost-Effective Carbon Capture Solution
- ② Why Carbon Capture Technology Is Failing
- ③ Trees vs. Whales: Which Absorbs More Oxygen?
- ④ The Mystery of Whale Falls in the Deep Sea
- ⑤ High-Tech Solutions for a Warmer Planet

다음 글을 읽고 필자가 전달하고자 하는 시사점으로 가장 적절한 것을 고르시오.

It is important to recognize that whales are valuable partners in our fight against the climate crisis and that protecting them is an effective and cost-efficient way to address this global challenge. Let's take action to help more whales live longer and healthier lives. Despite the important roles of whales in our environment, their lives are at serious risk. Today, whales are still threatened by various man-made dangers, such as fishing nets, ship strikes, and microplastic pollution in the ocean.

- ① Protecting whales is a practical and economical strategy for combating climate change.
- ② The international ban on whaling in 1986 has completely solved the problem of declining whale populations.
- ③ Advanced technology is the only reliable solution for reducing atmospheric CO₂.
- ④ Climate change has minimal effects on marine ecosystems compared to land ecosystems.
- ⑤ Individual actions have little impact on global environmental issues.

04 세부사항 · Details

다음 글의 내용과 일치하는 것을 고르시오.

Whales are amazing creatures. The largest blue whales can grow as long as 30 meters and weigh over 200 tons, and some whales have been known to live for more than 100 years. According to scientists, a single whale can absorb about 550 kilograms of CO₂ per year, while a tree can absorb only 22. Whale falls permanently bury about 145,000 tons of CO₂ under the sea every year. This achievement is basically what expensive carbon capture technologies aim to do—capture carbon and then store it underground.

- ① A single whale absorbs about 25 times more CO₂ per year than a tree.
- ② Blue whales can grow up to 200 meters long.
- ③ No whale has ever been known to live beyond 50 years.
- ④ Trees absorb more CO₂ per year than whales do.
- ⑤ Whale falls release CO₂ back into the atmosphere.

다음 글의 내용과 일치하지 않는 것을 고르시오.

Whales spread nutrients in the ocean not only vertically but also horizontally through a process called the great whale conveyor belt. In the summer, humpback whales tend to stay near Alaska, where the water is cold but rich in nutrients and prey for them to feed on. As winter approaches, they travel up to 4,800 kilometers to breed in the warm waters near Hawaii. Nutrients are relatively scarce in warm waters, which makes it difficult for phytoplankton to survive. Fortunately, when the whales arrive in Hawaii after eating their fill in Alaska, they release a large amount of nutrient-rich waste.

- ① Humpback whales stay near Alaska during the summer.
- ② Humpback whales travel up to 4,800 kilometers to breed.
- ③ Warm waters near Hawaii are rich in nutrients year-round.
- ④ The great whale conveyor belt moves nutrients horizontally.
- ⑤ Whales release nutrient-rich waste after eating their fill.

12 빈칸(구) 어려움

[6점]

다음 빈칸에 들어갈 말로 가장 적절한 것을 고르시오.

Whales dive deep into the ocean to feed on prey such as krill and small fish, and they occasionally swim up to the surface to breathe. When they near the surface, they release an enormous amount of waste. This waste is rich in nutrients that are essential for creatures living in that area. Specifically, it creates the perfect growing conditions for some microscopic creatures floating at the surface of the ocean. These creatures, called phytoplankton, absorb about 40% of the carbon in the atmosphere and produce about 50% of the atmosphere's oxygen. It is estimated that increasing the world's phytoplankton by 1% is equivalent to planting 2 billion new trees. In this way, whales indirectly contribute to reducing CO₂ by _____.

- ① providing nutrients that support phytoplankton growth
- ② directly absorbing carbon dioxide from the air
- ③ migrating to warmer waters during winter
- ④ sinking to the ocean floor after death
- ⑤ consuming large amounts of krill and fish

13 빈칸(문장) 어려움

[6점]

다음 빈칸에 들어갈 말로 가장 적절한 것을 고르시오.

Whales spread nutrients in the ocean not only vertically but also horizontally through a process called the great whale conveyor belt. In the summer, humpback whales tend to stay near Alaska, where the water is cold but rich in nutrients. As winter approaches, they travel up to 4,800 kilometers to breed in the warm waters near Hawaii. Nutrients are relatively scarce in warm waters. Fortunately, when the whales arrive in Hawaii after eating their fill in Alaska, they release a large amount of nutrient-rich waste. This means that the great whale conveyor belt essentially _____.

- ① transports nutrients from nutrient-rich cold waters to nutrient-poor warm waters
- ② prevents phytoplankton from growing in cold water regions
- ③ helps whales avoid predators during their long migration
- ④ increases the temperature of the ocean along migration routes
- ⑤ reduces the amount of oxygen available in tropical waters

14 함축 보통

[5점]

다음 글의 밑줄 친 'This achievement is basically what expensive carbon capture technologies aim to do'가 의미하는 바로 가장 적절한 것을 고르시오.

According to scientists, a single whale can absorb about 550 kilograms of CO₂ per year, while a tree can absorb only 22. Moreover, after they die, whales sink to the ocean floor with all the CO₂ they absorbed during their lives. Their bodies then become food and habitats for deep-sea creatures. This process is called a whale fall. Whale falls permanently bury about 145,000 tons of CO₂ under the sea every year. This achievement is basically what expensive carbon capture technologies aim to do—capture carbon and then store it underground.

- ① Whale falls naturally accomplish the same goal that costly carbon capture technologies pursue.
- ② Carbon capture technology was inspired by the study of whale falls.
- ③ Whales are more expensive to protect than developing new carbon capture systems.
- ④ The CO₂ buried by whale falls will eventually be released back into the atmosphere.
- ⑤ Scientists have abandoned carbon capture technology in favor of whale conservation.

15 순서배열 보통

[5점]

주어진 글 다음에 이어질 글의 순서로 가장 적절한 것을 고르시오.

Whales dive deep into the ocean to feed on prey such as krill and small fish, and they occasionally swim up to the surface to breathe. (A) These creatures, called phytoplankton, may look tiny and insignificant, but when taken together, they play a significant role in reducing CO₂. (B) When they near the surface, they release an enormous amount of waste. This waste is rich in nutrients that are essential for creatures living in that area. (C) Specifically, it creates the perfect growing conditions for some microscopic creatures floating at the surface of the ocean.

- ① (A)-(B)-(C)
- ② (A)-(C)-(B)
- ③ (B)-(C)-(A)
- ④ (B)-(A)-(C)
- ⑤ (C)-(A)-(B)

16 무관문장 쉬움

[3점]

다음 글에서 전체 흐름과 관계 없는 문장을 고르시오.

① Over the past two centuries, human activities have generated a significant amount of carbon dioxide. ② This gas itself is not dangerous when we encounter it in everyday situations. ③ However, excessive amounts of this gas in the atmosphere have contributed to the greenhouse effect. ④ Meanwhile, many deep-sea fish have developed bioluminescence to survive in complete darkness. ⑤ To make matters worse, this increase has sped up in recent years and is seriously threatening life on our planet.

- ① ①
- ② ②
- ③ ③
- ④ ④
- ⑤ ⑤

다음 글의 밑줄 친 'They'가 가리키는 대상으로, 'phytoplankton(식물성 플랑크톤)'이면 T, 'whales(고래)'이면 F를 고르시오.

These creatures, called phytoplankton, may look tiny and insignificant, but when taken together, they play a significant role in reducing CO₂. They absorb about 40% of the carbon in the atmosphere and produce about 50% of the atmosphere's oxygen. They do this through photosynthesis, a process that creates energy using carbon, water, and light.

- ① T
② F

D

05 서술형 · Writing

18 서술형 요약 보통

[5점]

다음 글을 읽고, 주어진 조건에 맞게 빈칸을 완성하십시오. [조건] 1. 아래 보기에서 적절한 단어를 골라 사용할 것 2. 필요시 형태를 변형할 것 [보기] absorb / bury / nutrient / pump / capture → Whales contribute to reducing CO₂ in two ways: first, whale falls permanently _____ CO₂ under the sea; second, whale _____ circulate _____(s) that support phytoplankton growth.

According to scientists, a single whale can absorb about 550 kilograms of CO₂ per year. After they die, whales sink to the ocean floor with all the CO₂ they absorbed during their lives. Whale falls permanently bury about 145,000 tons of CO₂ under the sea every year. In addition, whales make a significant indirect contribution to reducing CO₂ through an act called a whale pump. This refers to the vertical movements of whales that help circulate nutrients across different depths of the ocean.

▶ _____

19 서술형 영작 어려움

[7점]

다음 우리말과 같은 뜻이 되도록 주어진 조건에 맞게 영작하십시오. [우리말] 세계의 식물성 플랑크톤을 1% 늘리는 것은 20억 그루의 새 나무를 심는 것과 같다고 추정된다. [조건] 1. 아래 보기의 단어를 모두 사용할 것 (필요시 형태 변형 가능) 2. 「It is estimated that ~」 구문으로 시작할 것 [보기] increase / world / phytoplankton / 1% / equivalent / plant / 2 billion / new tree

These creatures, called phytoplankton, absorb about 40% of the carbon in the atmosphere and produce about 50% of the atmosphere's oxygen. It is estimated that increasing the world's phytoplankton by 1% is equivalent to planting 2 billion new trees.

▶ _____

20 서술형 어형 쉬움

[3점]

다음 글의 괄호 안의 단어를 문맥에 맞는 형태로 바꾸어 쓰시오. → Whale falls permanently bury about 145,000 tons of CO₂ under the sea every year. This achievement is basically what expensive carbon capture technologies aim (do). → 정답: _____

After they die, whales sink to the ocean floor with all the CO₂ they absorbed during their lives. Their bodies then become food and habitats for deep-sea creatures. This process is called a whale fall. Whale falls permanently bury about 145,000 tons of CO₂ under the sea every year. This achievement is basically what expensive carbon capture technologies aim to do.

▶ _____

1	2	3	4	5
③	①	①	①	③
6	7	8	9	10
③	①	①	①	①
11	12	13	14	15
③	①	①	①	③
16	17	18	19	20
④	①	bury, pumps, nutrient	It is estimated that increasing the world's phytoplankton by 1% is equivalent to planting 2 billion new trees.	to do

1 어법 ㉠~㉢ | 보통 | 5점 | 정답 ③

[해석] [원문] whales sink to the ocean floor with all the CO₂ they absorbed during their lives [해석] 고래는 살아있는 동안 흡수한 모든 CO₂와 함께 해저로 가라앉는다. 관계대명사 목적격 생략 구문에서 동사는 주절 시제(die → 과거)에 맞춰 과거형 absorbed가 되어야 하므로, 현재분사 absorbing은 어법상 어색하다.

[분석] ㉠ known: have been known to-v (…하는 것으로 알려져 있다) 현재완료 수동태 → known ✓ ㉡ beneficial: do beneficial things 형용사가 명사 things 수식 → beneficial ✓ ㉢ absorbing: 관계대명사 목적격 생략, 'CO₂ (that) they absorbed' → 과거형 absorbed가 맞음. absorbing ✕ → absorbed ✓ ㉣ is called: This process is called ~ 수동태 → is called ✓ ㉤ bury: Whale falls(복수 주어) + bury(복수 동사) → bury ✓

㉡ 관계대명사 목적격 생략 구문: 선행사 + (that/which) + S + V → V의 시제·태를 확인! 비교: absorbing(현재분사) vs. absorbed(과거분사/과거시제) — 여기서서는 during their lives(그들의 삶 동안)와 함께 쓰여 과거시제가 필요. have been known to-v ≈ be said to-v (…라고 알려져 있다)

2 어법 A/B/C | 어려움 | 6점 | 정답 ①

[해석] [원문] nutrients that are essential for creatures / microscopic creatures floating at the surface / when taken together [해석] 생물에게 필수적인 영양분 / 해수면에 떠다니는 미세 생물 / 함께 고려하면. (A) 선행사 nutrients + 주격 관계대명사 that, (B) 능동·진행의 현재분사 floating이 creatures 수식, (C) 분사구문에서 주어 they가 take의 대상이므로 수동 taken.

[분석] (A) that vs. what: 선행사 nutrients가 있으므로 관계대명사 that ✓ / what은 선행사를 포함하므로 ✕ (B) floating vs. floated: creatures가 '떠다니는(능동·진행)' 상태이므로 현재분사 floating ✓ / 과거분사 floated는 수동 의미로 부적절 ✕ (C) taking vs. taken: 분사구문 when (they are) taken together에서 주어 they(phytoplankton)가 take의 대상(수동) → taken ✓ / taking(능동)은 의미상 부적절 ✕

㉡ 관계대명사 선택: 선행사 있으면 that/which/who, 선행사 포함이면 what 현재분사(능동·진행) vs. 과거분사(수동·완료): creatures floating(떠다니는) / creatures floated(떠내려간?→✕) 분사구문 수동: when taken = when they are taken take together ≈ consider as a whole (전체로 고려하다)

3 어법 ㉠~㉣ | 어려움 | 6점 | 정답 ①

[해석] [원문] The number of whales had dramatically dropped until killing them for commercial purposes was officially banned in 1986. [해석] 고래의 수는 1986년에 상업적 목적의 포획이 공식적으로 금지될 때까지 급격히 감소했다. until절의 동사 was banned가 과거시제이므로, 주절도 과거시제 dropped이 자연스럽다. 과거완료 had dropped은 기준 과거 시점이 명시되어야 하는데, 여기서는 until 1986이 종료 시점을 나타내므로 단순과거가 적절하다.

[분석] ㉠ had dramatically dropped: until + 과거(was banned in 1986) → 주절은 단순과거 dropped이 자연스러움. had dropped(과거완료)은 '또 다른 과거 이전'을 나타내야 하는데, 여기서는 1986년까지의 감소를 기술하므로 과거시제가 적절 ✕ → dropped ✓ ㉡ fell: It is estimated that ~ fell 추정 내용의 과거 사실 서술 → fell ✓ ㉢ to recognize: It is important to-v 가주어·진주어 구문 → to recognize ✓ ㉣ protecting: protecting them is ~ 동명사 주어 → protecting ✓

㉡ 과거완료 vs. 단순과거: 과거완료(had p.p.)는 '과거의 과거'를 나타낼 때 사용. until + 과거시제와 함께 쓸 때는 단순과거가 자연스러운 경우가 많음. It is important to-v: 가주어(it) + 진주어(to-v) 구문 동명사 주어: Protecting them is ~ (동명사구가 주어 → 단수 취급)

4 어휘 A/B/C | 보통 | 5점 | 정답 ①

[해석] [원문] have generated a significant amount of CO₂ / resulting in a gradual increase / seriously threatening life [해석] 상당한 양의 CO₂를 생성했다 / 점진적인 온도 상승을 초래하며 / 생명을 심각하게 위협하고 있다. (A) 인간 활동이 많은 CO₂를 만들었다는 맥락 → significant(상당한), (B) 온실효과로 인한 '점진적' 상승 → gradual, (C) 기후변화가 생명을 '위협'한다 → threatening.

[분석] (A) significant(상당한) vs. insignificant(미미한): 온실효과를 유발할 만큼 '많은' 양이므로 significant ✓ / insignificant는 문맥과 반대 ✕ (B) gradual(점진적인) vs. sudden(갑작스러운): 원문에서 '점진적 증가'이며 최근 가속되었다고 했으므로 gradual ✓ / sudden은 원문과 불일치 ✕ (C) threatening(위협하는) vs. protecting(보호하는): 기후변화가 생명을 '위협'하는 맥락 → threatening ✓ / protecting은 정반대 ✕

㉡ significant ≈ considerable, substantial (상당한) ↔ insignificant ≈ negligible (미미한) gradual ≈ slow, steady (점진적인) ↔ sudden ≈ abrupt (갑작스러운) threaten ≈ endanger, jeopardize (위협하다) ↔ protect ≈ safeguard (보호하다) 문맥 흐름: CO₂ 많이 생성 → 온실효과 → 점진적 온도 상승 → 생명 위협

5 어휘 부적절 | 어려움 | 6점 | 정답 ③

[해석] [원문] may look tiny and insignificant [해석] 아주 작고 하찮아 보일 수 있다. 원문은 insignificant(하찮은)인데 ㉢에서 significant(중요한)로 바뀌어 있다. 문맥상 '작고 하찮아 보이지만(but) 중요한 역할을 한다'는 역접 구조이므로, ㉢은 insignificant가 되어야 한다.

[분석] ㉠ essential: 영양분이 생물에게 '필수적인' → essential ✓ ㉡ microscopic: '현미경으로 봐야 할 정도로 작은' 생물 → microscopic ✓ ㉢ significant: 원문은 insignificant(하찮은). tiny and insignificant(작고 하찮은)이 but 이하의 significant role과 대조 → significant ✕ → insignificant ✓ ㉣ significant: play a significant role(중요한 역할을 하다) → significant ✓ ㉤ light: 광합성에 탄소, 물, 빛을 사용 → light ✓

㉡ insignificant ≈ trivial, negligible (하찮은) ↔ significant ≈ important, considerable (중요한) 역접(but) 앞뒤 의미 대조 패턴: tiny and insignificant ↔ play a significant role microscopic ≈ extremely small, minute (극히 작은) essential ≈ vital, indispensable, crucial (필수적인)

6 어휘 부적절 | 보통 | 5점 | 정답 ③

[해석] [원문] Nutrients are relatively scarce in warm waters [해석] 따뜻한 물에서는 영양분이 비교적 부족하다. 원문은 scarce(부족한)인데 ㉢에서 abundant(풍부한)로 바뀌어 있다. 따뜻한 물에서 영양분이 부족하기 때문에 식물성 플랑크톤이 살기 어렵다는 맥락이므로, abundant는 부적절하다.

[분석] ㉠ horizontally: 수직적(vertically)뿐 아니라 수평적(horizontally)으로도 → horizontally ✓ ㉡ migrate: 거대한 포유류가 장거리를 '이동하다' → migrate ✓ ㉢ abundant: 원문은 scarce(부족한). 따뜻한 물에서 영양분이 부족해서 식물성 플랑크톤이 살기 어렵다는 문맥 → abundant(풍부한) ✕ → scarce ✓ ㉣ nutrient-rich: '영양분이 풍부한' 배설물 → nutrient-rich ✓ ㉤ encourage: 성장을 '촉진하다' → encourage ✓

㉡ scarce ≈ rare, insufficient (부족한, 드문) ↔ abundant ≈ plentiful, ample (풍부한) migrate ≈ move, travel seasonally (이동하다) nutrient-rich ≈ nourishing (영양분이 풍부한) encourage ≈ promote, stimulate, foster (촉진하다) 인과 관계 확인: 영양분 부족(scarce) → 생존 어려움(difficult to survive)

7 주제 | 보통 | 5점 | 정답 ①

[해석] [원문] whales also make a significant indirect contribution to reducing CO₂ through an act called a whale pump [해석] 고래는 고래 펌프라 불리는 행동을 통해 CO₂ 감소에 상당한 간접적 기여를 한다. 지문은 고래 펌프가 영양분을 순환시켜 식물성 플랑크톤의 성장을 돕고, 이것이 CO₂ 흡수를 이어진다는 내용이다.

[분석] ㉠ the role of whale pumps in reducing CO₂ through nutrient circulation: 고래 펌프가 영양분 순환을 통해 CO₂를 줄이는 역할 → 지문의 핵심 내용과 정확히 일치 ✓ ㉡ how whales absorb CO₂ directly from the atmosphere: 직접 흡수가 아닌 '간접적 기여'가 핵심 → 부분적, 부정확 ✕ ㉢ the importance of phytoplankton in deep-sea ecosystems: 식물성 플랑크톤은 해수면(surface)에 서식, deep-sea가 아님 ✕ ㉣ various methods to protect endangered whale species: 고래 보호 방법은 이 지문에서 다루지 않음 ✕ ㉤ the harmful effects of CO₂ on marine life: CO₂가 해양 생물에 미치는 피해는 이 지문의 주제가 아님 ✕

㉡ 주제 파악 전략: 반복 키워드(whale pump, nutrients, CO₂, phytoplankton) + 첫 문장의 핵심어(indirect contribution to reducing CO₂) 확인 whale pump = 고래의 수직 이동으로 인한 영양분 순환 메커니즘 indirect contribution ≈ 간접적 기여

8 제목 | 보통 | 5점 | 정답 ①

[해석] [원문] there are also low-tech, economical, and even long-lasting solutions available / This achievement is basically what expensive carbon capture technologies aim to do [해석] 저비용이면서 오래 지속되는 해결책이 있다 / 이 성과는 기본적으로 비싼 탄소 포집 기술이 목표로 하는 것이다. 고비용 기술과 대비하여 고래가 경제적인 자연 기반 탄소 흡수원이라는 것이 핵심.

[분석] ① Whales: Nature's Cost-Effective Carbon Capture Solution: 고래가 자연의 비용 효율적 탄소 포집 해결책이라는 지문 핵심과 정확히 부합 ✓ ② Why Carbon Capture Technology Is Failing: 탄소 포집 기술의 실패가 아닌 고비용 대비 고래의 장점이 핵심 ✕ ③ Trees vs. Whales: Which Absorbs More Oxygen?: 산소가 아닌 CO₂ 흡수 비교이며, 이것은 전체가 아닌 부분적 내용 ✕ ④ The Mystery of Whale Falls in the Deep Sea: whale fall은 소재 중 하나이지 신비(mystery)가 주제가 아님 ✕ ⑤ High-Tech Solutions for a Warmer Planet: 지문은 high-tech가 아닌 low-tech 해결책을 강조 ✕

➡ 제목 = 글의 핵심 주제를 함축적으로 표현 cost-effective ≈ economical ≈ affordable (비용 효율적인) nature-based solution (자연 기반 해결책) — 핵심 대조: high-tech & expensive ↔ low-tech & economical carbon capture ≈ 탄소 포집 기술

9 시사점 | 어려움 | 6점 | 정답 ①

[해석] [원문] protecting them is an effective and cost-efficient way to address this global challenge [해석] 고래를 보호하는 것은 이 전 지구적 과제를 해결하는 효과적이고 비용 효율적인 방법이다. 필자는 고래 보호가 기후 위기에 대한 실질적이고 경제적인 전략임을 강조하며 행동을 촉구하고 있다.

[분석] ① Protecting whales is a practical and economical strategy for combating climate change: 필자의 핵심 메시지와 정확히 일치 — effective, cost-efficient → practical, economical ✓ ② The international ban on whaling in 1986 has completely solved the problem: 'completely solved'는 과장. 지문에서 여전히 위험받고 있다고 언급 ✕ ③ Advanced technology is the only reliable solution: 지문은 자연 기반 해결책(고래)을 강조, high-tech only가 아님 ✕ ④ Climate change has minimal effects on marine ecosystems: 지문 내용과 무관하며 오히려 반대 ✕ ⑤ Individual actions have little impact: 지문은 'Let's take action'이라며 행동을 촉구 → 반대 취지 ✕

➡ 시사점 = 필자가 글 전체를 통해 궁극적으로 전달하려는 메시지 practical ≈ effective ≈ workable (실질적인) economical ≈ cost-efficient ≈ affordable (경제적인) combat ≈ fight against ≈ tackle (맞서 싸우다) 필자 태도 키워드: 'Let's take action' → 행동 촉구

10 일치 | 쉬움 | 3점 | 정답 ①

[해석] [원문] a single whale can absorb about 550 kilograms of CO₂ per year, while a tree can absorb only 22 [해석] 한 마리 고래는 연간 약 550kg의 CO₂를 흡수할 수 있는 반면, 나무 한 그루는 22kg만 흡수할 수 있다. 550÷22=25로, 고래가 나무보다 약 25배 더 많은 CO₂를 흡수한다.

[분석] ① A single whale absorbs about 25 times more CO₂ per year than a tree: 550÷22=25 → 약 25배 ✓ ② Blue whales can grow up to 200 meters: 30 meters가 맞음, 200은 무게(톤) ✕ ③ No whale has ever been known to live beyond 50 years: 100년 이상 사는 고래가 있다고 함 ✕ ④ Trees absorb more CO₂ than whales: 나무 22kg < 고래 550kg → 반대 ✕ ⑤ Whale falls release CO₂ back into the atmosphere: permanently bury(영구히 묻는다)이므로 방출이 아님 ✕

➡ 수치 비교 문제: 550kg(고래) vs. 22kg(나무) → 550÷22≈25배 일치 문제 풀이 전략: 선지의 숫자·단위·비교 방향을 원문과 정확히 대조 permanently bury ≈ store forever (영구히 저장하다) ↔ release ≈ emit (방출하다)

11 불일치 | 보통 | 5점 | 정답 ③

[해석] [원문] Nutrients are relatively scarce in warm waters [해석] 따뜻한 물에서는 영양분이 비교적 부족하다. ③번은 '하와이 근처 따뜻한 물은 연중 영양분이 풍부하다'라고 했는데, 원문에서는 영양분이 부족(scarce)하다고 했으므로 불일치한다.

[분석] ① Humpback whales stay near Alaska during the summer: 원문 'In the summer, humpback whales tend to stay near Alaska' 일치 ✓ ② Humpback whales travel up to 4,800 km to breed: 원문 'they travel up to 4,800 kilometers to breed' 일치 ✓ ③ Warm waters near Hawaii are rich in nutrients year-round: 원문 'Nutrients are relatively scarce in warm waters' → rich가 아닌 scarce이므로 불일치 ✕ (정답) ④ The great whale conveyor belt moves nutrients horizontally: 원문 'horizontally through a process called the great whale conveyor belt' 일치 ✓ ⑤ Whales release nutrient-rich waste after eating their fill: 원문 일치 ✓

➡ 불일치 문제 = 원문과 다른 내용 찾기 scarce(부족함) ↔ rich/abundant(풍부함) — 반의어 함정에 주의! year-round(연중) = 원문에 없는 추가 정보 삽입 → 불일치 근거 tend to stay ≈ usually remain (보통 ~에 머무르다)

12 빈칸(구) | 어려움 | 6점 | 정답 ①

[해석] [원문] This waste is rich in nutrients ... it creates the perfect growing conditions for ... phytoplankton [해석] 이 배설물은 영양분이 풍부하며 식물성 플랑크톤의 완벽한 성장 조건을 만든다. 고래가 간접적으로 CO₂ 감소에 기여하는 방식은 '영양분을 제공하여 식물성 플랑크톤 성장을 돕는 것'이다.

[분석] ① providing nutrients that support phytoplankton growth: 고래 배설물 → 영양분 → 식물성 플랑크톤 성장 → CO₂ 흡수 연결 고리와 정확히 일치 ✓ ② directly absorbing carbon dioxide from the air: 직접 흡수가 아닌 '간접적' 기여가 핵심 ✕ ③ migrating to warmer waters during winter: 이동은 great whale conveyor belt 관련, 이 지문의 핵심이 아님 ✕ ④ sinking to the ocean floor after death: whale fall은 직접적 기여, 여기서는 간접적 기여를 설명 ✕ ⑤ consuming large amounts of krill and fish: 먹이 섭취는 과정의 일부이지 CO₂ 감소 기여 방식 자체가 아님 ✕

➡ 빈칸(구) 전략: 빈칸 앞의 'by' = 수단/방법 → '어떤 방법으로' 간접 기여하는지 찾기 논리 흐름: 고래 배설물(waste) → 영양분(nutrients) → 식물성 플랑크톤(phytoplankton) → CO₂ 흡수 indirectly ≈ 간접적으로 ↔ directly ≈ 직접적으로 provide ≈ supply, deliver (제공하다)

13 빈칸(문장) | 어려움 | 6점 | 정답 ①

[해석] [원문] This process can be thought of as a conveyor belt that moves nutrients from cold waters to warm waters [해석] 이 과정은 차가운 물에서 따뜻한 물로 영양분을 이동시키는 컨베이어 벨트로 생각할 수 있다. 고래가 알래스카(영양분 풍부한 찬 물)에서 먹고 하와이(영양분 부족한 따뜻한 물)로 이동하며 영양분을 운반하는 과정이다.

[분석] ① transports nutrients from nutrient-rich cold waters to nutrient-poor warm waters: 알래스카(cold, nutrient-rich) → 하와이(warm, nutrient-scarce)로 영양분 운반 → 정확히 일치 ✓ ② prevents phytoplankton from growing in cold water regions: 오히려 성장을 촉진, 방해가 아님 ✕ ③ helps whales avoid predators during their long migration: 포식자 회피는 지문에 언급 없음 ✕ ④ increases the temperature of the ocean: 수온 변화는 지문 내용과 무관 ✕ ⑤ reduces the amount of oxygen available in tropical waters: 산소 감소가 아닌 영양분 공급이 핵심 ✕

➡ 빈칸(문장) 전략: 빈칸 앞의 'essentially'(본질적으로)와 'This means that'(이것은 ~을 의미한다) → 앞 내용의 핵심 요약 conveyor belt ≈ 컨베이어 벨트(자동 운반 장치) — 비유적 표현 transport ≈ move, carry, deliver (운반하다) nutrient-rich ↔ nutrient-poor (영양분 풍부한 ↔ 영양분 부족한)

14 함축 | 보통 | 5점 | 정답 ①

[해석] [원문] This achievement is basically what expensive carbon capture technologies aim to do—capture carbon and then store it underground [해석] 이 성과는 기본적으로 비싼 탄소 포집 기술이 목표로 하는 것, 즉 탄소를 포집하여 지하에 저장하는 것이다. 고래 낙하(whale fall)가 자연적으로 비싼 기술과 같은 목표를 달성한다는 뜻이다.

[분석] ① Whale falls naturally accomplish the same goal that costly carbon capture technologies pursue: whale fall이 비싼 기술과 동일한 목표(탄소 포집·저장)를 자연적으로 달성 → 정확한 함축 ✓ ② Carbon capture technology was inspired by whale falls: 영감 관계는 언급 없음 ✕ ③ Whales are more expensive to protect: 지문은 고래가 '경제적' 해결책이라고 강조 → 반대 ✕ ④ CO₂ buried by whale falls will eventually be released: permanently(영구적으로) bury → 방출되지 않음 ✕ ⑤ Scientists have abandoned carbon capture technology: 포기했다는 내용 없음 ✕

➡ 함축 의미 파악: 밑줄 문장의 핵심 = 'This achievement(whale fall의 CO₂ 매장) = expensive carbon capture(비싼 기술의 목표)' basically ≈ essentially, fundamentally (기본적으로) aim to do ≈ seek to accomplish, pursue (목표로 하다) accomplish ≈ achieve ≈ fulfill (달성하다)

15 순서배열 | 보통 | 5점 | 정답 ③

[해석] [해석] 주어진 문장: 고래가 먹이를 먹으려 깊이 잠수하고, 때때로 숨을 쉬기 위해 수면으로 올라온다. (B) 수면 가까이에서 엄청난 양의 배설물을 방출한다. 이 배설물은 영양분이 풍부하다. (C) 구체적으로, 해수면에 떠다니는 미세 생물의 완벽한 성장 조건을 만든다. (A) 이 생물을 식물성 플랑크톤이라 부르며, CO₂ 감소에 중요한 역할을 한다. → B(배설물: 영양분) → C(구체적으로: 미세 생물 성장 조건) → A(이 생물 = 식물성 플랑크톤의 역할)

[분석] 정답 (B)-(C)-(A) ✓ (B)가 먼저: 주어진 문장의 'swim up to the surface'에 이어 'When they near the surface' 연결 (C)가 다음: 'Specifically'(구체적으로)가 (B)의 'nutrients'를 구체화, 'it'은 waste를 지칭 (A)가 마지막: 'These creatures'가 (C)의 'microscopic creatures'를 지칭 나머지 선지: 지시어·접속부사 연결이 자연스럽지 않음 ✕

➡ 순서배열 전략: 1) 지시어 추적: These creatures(A) → microscopic creatures(C)를 받음 → C→A 2) 접속부사: Specifically(구체적으로) → 앞 내용을 구체화 → B→C 3) 대명사: it(C) = waste(B) → B→C 4) 논리 흐름: 배설물 방출 → 영양분 → 성장 조건 → 역할

16 무관문장 | 쉬움 | 3점 | 정답 ④

[해석] [원문] ①~③, ⑤는 CO₂와 온실효과로 인한 기후 위기에 대한 내용이다. ④ '심해 물고기가 완전한 어둠 속에서 생존하기 위해 생물발광 능력을 발달시켰다'는 심해 생물의 특성에 관한 내용으로, CO₂·온실효과·기후변화와 무관하다.

[분석] ① CO₂ 대량 생성 → 전체 흐름(CO₂ 문제)과 일치 ✓ ② CO₂ 자체는 일상에서 위험하지 않음 → CO₂ 설명 흐름 유지 ✓ ③ 대기 중 과도한 CO₂가 온실효과에 기여 → 핵심 논지 ✓ ④ 심해 물고기의 생물발광 → CO₂·온실효과와 완전히 무관 ✕ (정답) ⑤ 기온 상승이 가속화되어 생명 위협 → 핵심 논지 ✓

☛ 무관문장 전략: 글의 중심 화제(CO₂ → 온실효과 → 기후 위기)를 파악하고, 이탈하는 문장 찾기 bioluminescence = 생물발광 (심해 생물 관련 → 기후변화와 무관) Meanwhile(한편)는 화제 전환 신호 → 무관문장 단서 To make matters worse ≈ Even worse (설상가상으로) — 흐름 강화 신호

17 TF | 쉬움 | 3점 | 정답 ①

[해석] [원문] They absorb about 40% of the carbon in the atmosphere [해석] 그것들은 대기 중 탄소의 약 40%를 흡수한다. 밑줄 친 They의 선행 문맥은 'These creatures, called phytoplankton'이며, 광합성(photosynthesis)을 통해 탄소를 흡수하는 주체는 식물성 플랑크톤이다.

[분석] T(phytoplankton): 바로 앞 문장의 주어 'These creatures, called phytoplankton'을 받아 They로 지칭. 광합성으로 탄소를 흡수하고 산소를 생산하는 것은 식물성 플랑크톤의 기능 ✓ F(whales): 고래는 광합성을 하지 않으며, 문맥상 주어가 phytoplankton에서 전환되지 않았음 ✕

☛ 지칭 파악 전략: 대명사(They) → 가장 가까운 선행 명사(These creatures = phytoplankton) 확인 결정적 단서: photosynthesis(광합성) → 식물성 플랑크톤의 고유 기능 phytoplankton = phyto(식물) + plankton(플랑크톤) → 광합성 가능

18 서술형 요약 | 보통 | 5점 | 정답 bury, pumps, nutrient

[해석] [원문] Whale falls permanently bury about 145,000 tons of CO₂ / through an act called a whale pump / circulate nutrients [해석] 고래 낙하는 영구적으로 약 145,000톤의 CO₂를 매장한다 / 고래 펌프라 불리는 행동을 통해 / 영양분을 순환시킨다. 첫 번째 빈칸: bury(매장하다), 두 번째 빈칸: pumps(펌프), 세 번째 빈칸: nutrient(s)(영양분).

[분석] 첫 번째 빈칸 bury: 원문 'permanently bury about 145,000 tons of CO₂' → whale falls가 주어(복수), 현재시제 → bury ✓ 두 번째 빈칸 pumps: 원문 'whale pump' → whale pumps(고래 펌프) ✓ 세 번째 빈칸 nutrient(s): 원문 'circulate nutrients' → nutrient(s) ✓ 오답 가능성: absorb(흡수하다)와 capture(포집하다)는 문맥상 부적합

☛ 요약형 서술: 핵심 키워드를 원문에서 정확히 추출 bury ≈ store underground (지하에 저장하다) pump: 여기서는 명사로 '펌프' (고래의 수직 이동 메커니즘) nutrient ≈ nourishment (영양분) circulate ≈ distribute, spread (순환시키다)

19 서술형 영작 | 어려움 | 7점 | 정답 It is estimated that increasing the world's phytoplankton by 1% is equivalent to planting 2 billion new trees.

[해석] [원문] It is estimated that increasing the world's phytoplankton by 1% is equivalent to planting 2 billion new trees. [해석] 세계의 식물성 플랑크톤을 1% 늘리는 것은 20억 그루의 새 나무를 심는 것과 같다고 추정된다. 구문 분석: It(가주어) is estimated that(진주어절) increasing(동명사 주어) the world's phytoplankton by 1% is equivalent to planting(전치사 to + 동명사) 2 billion new trees.

[분석] 핵심 구문: 1) It is estimated that ~ : 가주어-진주어 구문, '~라고 추정된다' ✓ 2) increasing ~ by 1%: 동명사 주어 + by(차이/증감) ✓ 3) is equivalent to planting: be equivalent to + 동명사(전치사 to 뒤) ✓ 4) 2 billion new trees: 숫자 표현 + 복수 명사 ✓

☛ be equivalent to + V-ing: ~하는 것과 동등하다 (to는 전치사 → 뒤에 동명사!) It is estimated that ~ ≈ It is believed/thought that ~ (~라고 추정/여겨진다) increase ~ by N%: ~을 N% 늘리다 (by = 차이·증감폭) 주의: equivalent to planting (○) / equivalent to plant (✕) — to가 전치사이므로 동명사 필수

20 서술형 어형 | 쉬움 | 3점 | 정답 to do

[해석] [원문] what expensive carbon capture technologies aim to do [해석] 비싼 탄소 포집 기술이 목표로 하는 것. aim to-v 구문에서 aim 뒤에 to 부정사가 와야 하므로 (do) → to do가 정답이다.

[분석] aim to-v: '~하는 것을 목표로 하다' → aim to do ✓ aim doing: 문법적으로 부적절 ✕ aim do: to 없이는 성립 불가 ✕

☛ aim to-v (to 부정사를 목적어로 취하는 동사) 비슷한 패턴: want to-v, hope to-v, plan to-v, decide to-v 비교: enjoy v-ing, finish v-ing (동명사를 목적어로 취하는 동사) what ~ aim to do = the thing that ~ aim to do (관계대명사 what)